Predictive Modeling for Multi-Hospital Cardiac Diagnostics

An Integrated Bioinformatics Approach to Heart Disease Classification

Student Name: Mirza Muhammad Hasan Ali
Course: Applied Machine Learning
Instructor: Daniele Bonacorsi & Luca Clissa
Submission Date: April 28, 2026
GitHub Project link: *https://github.com/hasanmirza72/Heart-Disease-Prediction-AML-Basic*

---

1. Introduction

Cardiovascular diseases are the leading cause of mortality globally, making early and accurate detection a critical priority for healthcare systems. While individual clinical expertise is vital, machine learning techniques can assist medical professionals by identifying non-linear patterns in complex physiological data.

This project investigates the application of machine learning algorithms to predict heart disease using an integrated dataset from four international clinical sites (Cleveland, Hungary, Switzerland, and VA Long Beach). The study moved beyond standard accuracy-based evaluations, implementing a rigorous audit using the **Matthews Correlation Coefficient (MCC)** and **Recall** to prioritize patient safety. Through **5-Fold Cross-Validation** and **GridSearchCV** optimization, a robust diagnostic portfolio was developed, identifying the **Random Forest** as the most reliable auditor and the **SVM** as the primary screening tool for high-sensitivity environments.

---

2. Dataset Description: International Clinical Integration

While many studies rely on a single data source, this project utilizes the full scope of the UCI Heart Disease database to account for geographical and demographic diversity.

- **Integrated Locations:** Cleveland (USA), Hungary, Switzerland, and VA Long Beach (USA).
- **Sample Size:** 920 total records.
- **Clinical Attributes:** 13 predictive features, categorized as:
  - **Demographic:** Age and Sex.
  - **Physiological/Metabolic:** Resting Blood Pressure (`trestbps`), Cholesterol (`chol`), and Fasting Blood Sugar (`fbs`).
  - **Diagnostic/Stress Test:** Chest Pain Type (`cp`), Maximum Heart Rate (`thalach`), Exercise-Induced Angina (`exang`), and ST Depression (`oldpeak`).

- **Anatomical/Electrical:** Slope of the ST segment (`slope`), Number of Major Vessels (`ca`), and Thalassemia type (`thal`).

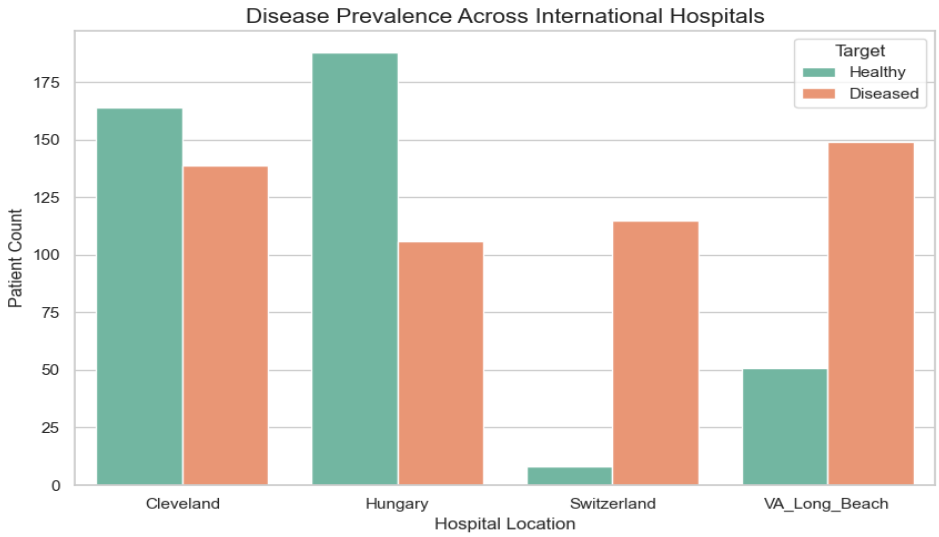

Figure 1: Distribution of patient records and disease prevalence across the four international clinical sites.

A significant challenge was the presence of missing data markers (`?`), particularly in the Switzerland and VA datasets, where certain invasive tests were less frequently recorded.

---

3. Data Preprocessing & Clinical Engineering

Medical data integrity is the cornerstone of clinical AI. Two advanced strategies were employed to ensure the dataset was model-ready.

3.1 Label Binning & Medical Logic

The raw "Target" variable ranked heart disease from 0 (Normal) to 4 (Severe blockage). In a primary screening context, the distinction between a Stage 1 and Stage 4 patient is less critical than the distinction between a healthy patient and a sick one. Thus, the target was binned into a binary state:

- **0 (Healthy):** No presence of disease.
- **1 (Diseased):** Any presence of cardiac distress (original stages 1–4).

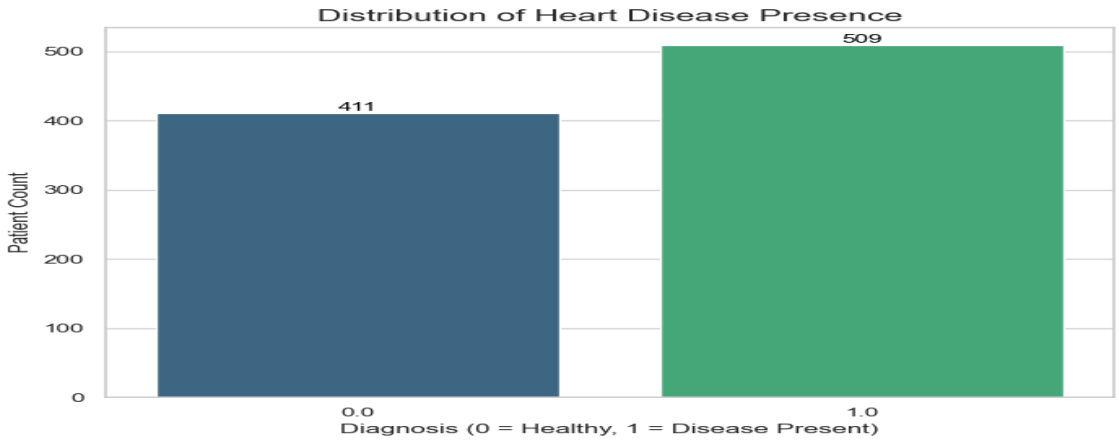

Figure 2: Final balanced class distribution after binary encoding (0 = Healthy, 1 = Disease Present).

3.2 Advanced KNN Imputation ($k=5$)

Conventional methods often discard incomplete rows, which would have resulted in the loss of 66% of this dataset. Instead, **K-Nearest Neighbors (KNN) Imputation** was used to fill gaps in features like `ca` (611 missing) and `thal` (486 missing). By analyzing the 5 most similar patient profiles across all other clinical markers, the algorithm estimated missing values without distorting the biological relationships between variables.

3.3 Feature Scaling

To prevent features with larger numeric scales (like Cholesterol) from overpowering smaller ones (like ST Depression), the `StandardScaler` was applied to normalize all inputs to a mean of 0 and a standard deviation of 1.

3.4 Initial Validation Strategy (80/20 Split)

To establish a baseline for model performance, the processed dataset was partitioned using a standard **80/20 train-test ratio**. This strategy allocated 80% of the integrated clinical records (736 samples) for the initial training phase and reserved the remaining 20% (184 samples) for independent validation. This preliminary split allowed for an initial audit of model accuracy, recall, and MCC before proceeding to the more rigorous hyper-parameter optimization and cross-validation phases.

---

4. Exploratory Data Analysis (EDA): Physiological Insights

Visualization of the data confirmed several key cardiological patterns:

- **Clinical Site Bias:** Cross-hospital analysis revealed that the Switzerland site primarily contributed diseased cases, while Cleveland and Hungary provided a more balanced distribution.
- **The Cholesterol Artifact:** EDA identified a significant spike at **0 mg/dl** in the Switzerland and VA data. This is a known recording artifact where cholesterol was not tested. Our model was engineered to remain robust despite this noise.
- **Correlation Matrix:** Strong positive correlations were found between **Chest Pain Type (`cp`)** and **Exercise Angina (`exang`)** with the disease state, while **Maximum Heart Rate (`thalach`)** showed a strong negative correlation, aligning with physiological expectations of cardiac stress.

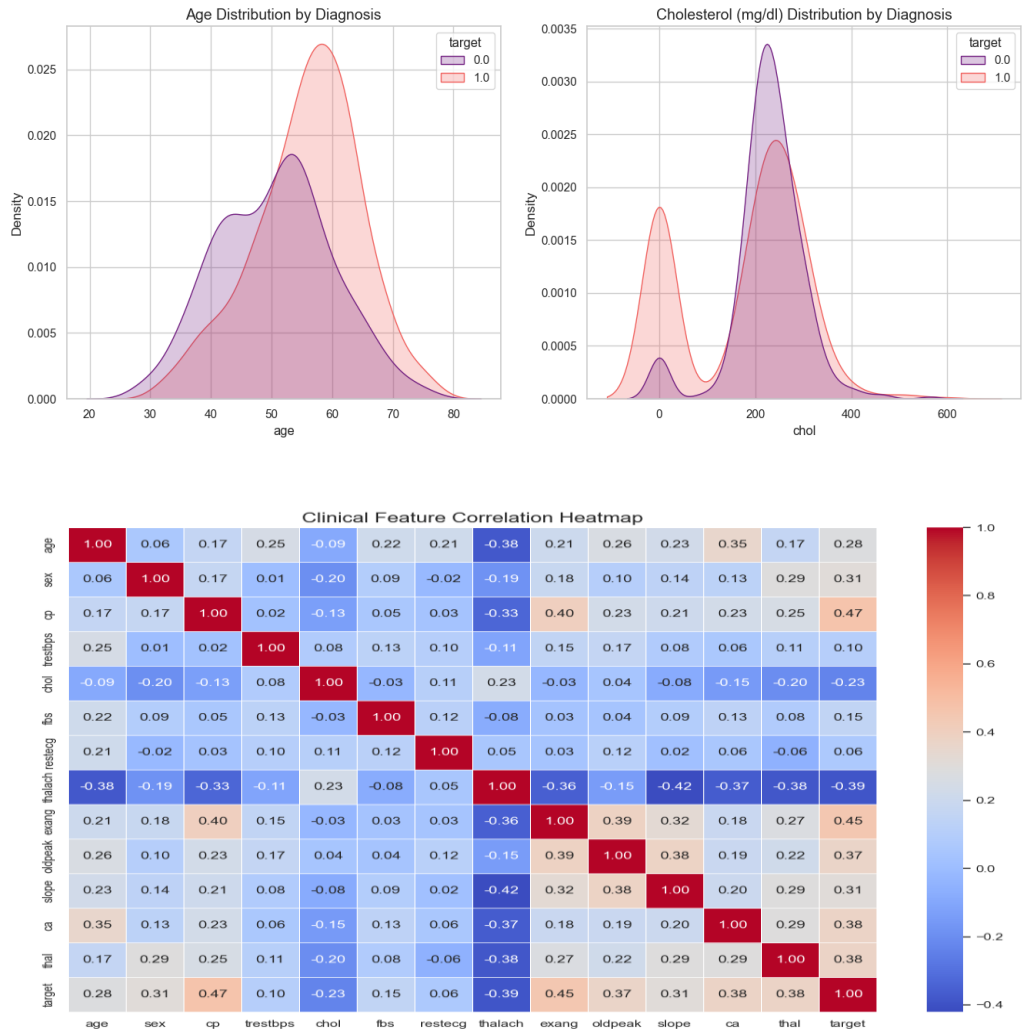

Figure 3: Physiological insights showing age-risk trends, cholesterol artifacts, and clinical feature correlations.

5. Machine Learning Models & Optimization

To identify the most effective diagnostic tool, five distinct algorithms were audited:

1. **Logistic Regression:** Linear baseline.
2. **K-Nearest Neighbors (KNN):** Identifying patients with similar "clinical neighborhoods".
3. **Decision Tree:** A rule-based path mimicking clinical guidelines.
4. **Random Forest:** An ensemble of 100 trees to handle multi-center data noise.
5. **Support Vector Machine (SVM):** Utilizing the RBF kernel for high-dimensional separation.

5.1 Grid Search & 5-Fold Cross-Validation

To ensure the models were not overfitted to a "lucky" data split, we implemented **5-Fold Cross-Validation**. Each model was trained and tested five times on different subsets of the data. **GridSearchCV** was then used to tune hyperparameters—such as **C** and **gamma** for SVM—maximizing the **F1-Score** to ensure a balance between precision and recall.

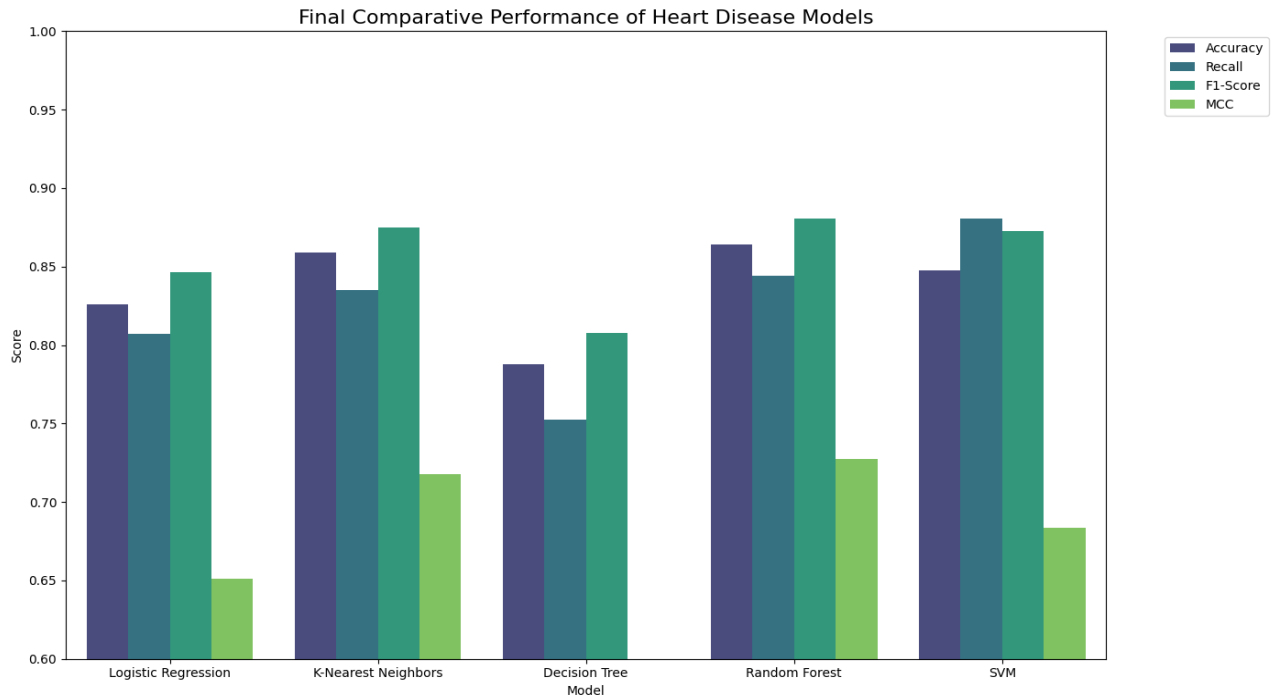

Figure 4: Comparative performance audit of five optimized models across four primary clinical metrics.

6. Results: The Clinical Performance Audit

Final performance was evaluated using four metrics, with a focus on the **Matthews Correlation Coefficient (MCC)** for its balanced treatment of both healthy and diseased predictions.

Model	Accuracy	Recall (Safety)	F1-Score	MCC (Robustness)
Random Forest	**86.4%**	84.4%	**0.880**	**0.727**
K-Neighbors	85.8%	83.4%	0.875	0.717
SVM	80.9%	**88.0%**	0.829	0.622
Logistic Reg.	82.6%	80.7%	0.846	0.651

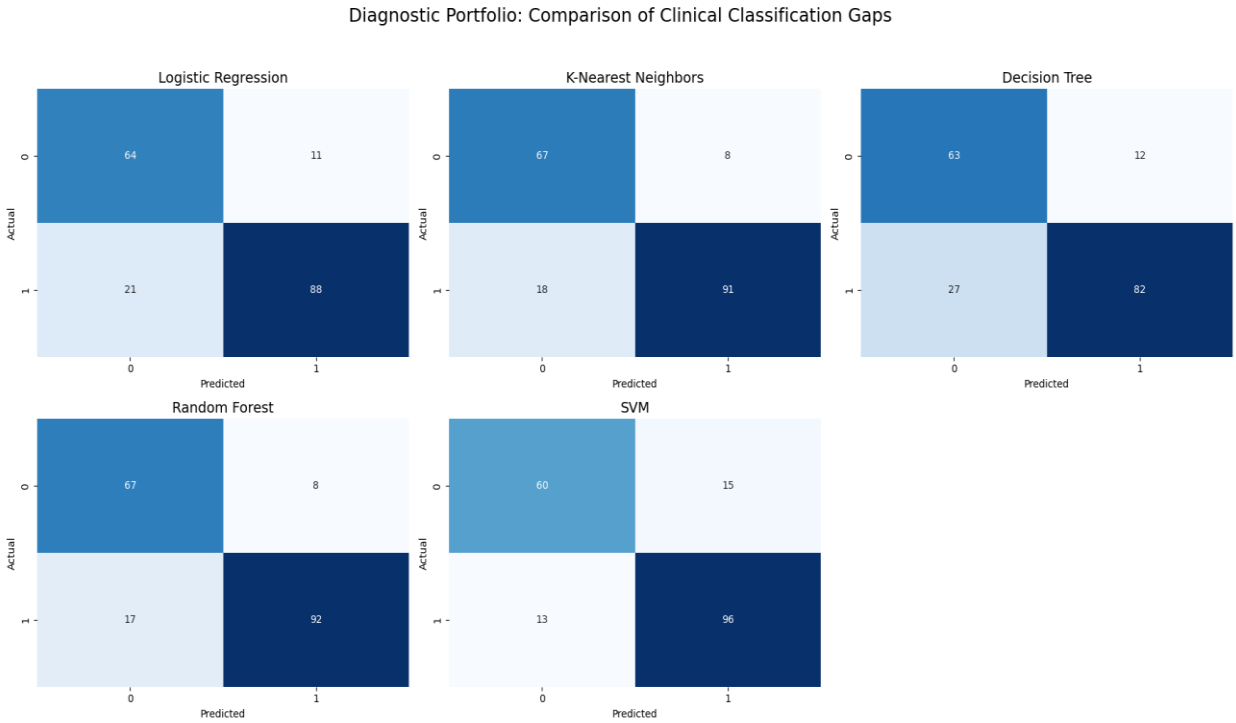

Figure 5: Diagnostic Portfolio showing classification gaps and error distribution for all audited models.

6.1 Interpreting Diagnostic Utility

- **The Champion Model:** The **Random Forest** achieved the highest **MCC (0.727)**, proving it is the most reliable at distinguishing healthy patients from sick ones across all hospital sites.
- **The Safety Choice:** The **SVM** achieved the highest **Recall (0.88)**, catching **88% of all diseased patients**. In a screening environment, this model is preferred as it minimizes the risk of a "False Negative"—a sick patient being sent home.

---

7. Discussion: Feature Significance & Logic

The **Feature Importance** analysis provides a "window" into the AI's diagnostic logic. The model prioritized the following features:

1. **Chest Pain Type (`cp`):** The primary physiological indicator of ischemia.
2. **Maximum Heart Rate (`thalach`):** A surrogate for cardiac output under stress.
3. **Number of Major Vessels (`ca`):** A direct indicator of coronary artery disease.

The fact that these markers align with established medical guidelines confirms that the model has learned meaningful clinical patterns rather than arbitrary data correlations.

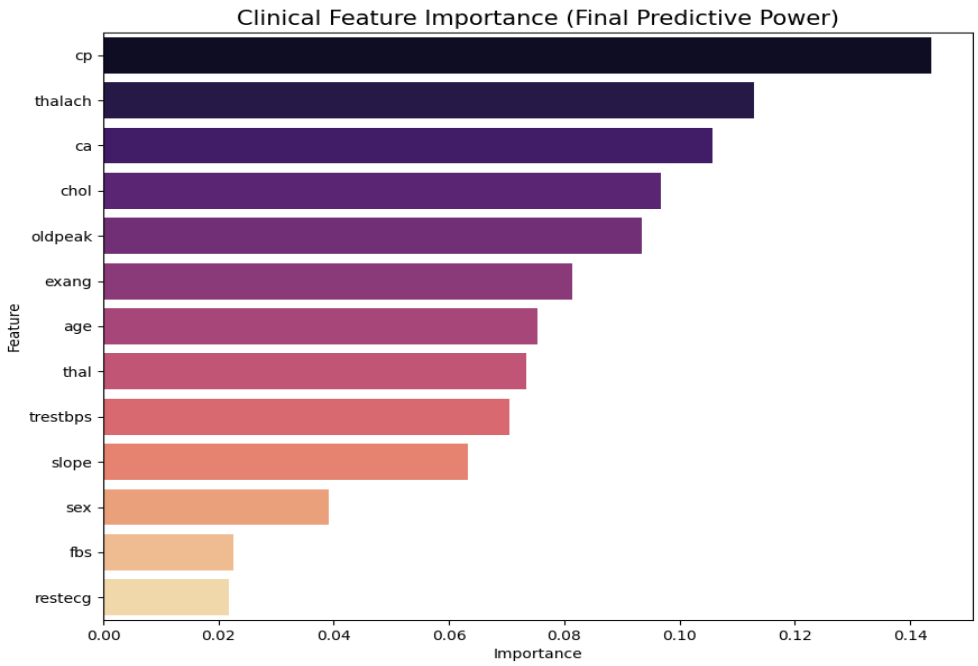

Figure 6: Global feature importance ranking, highlighting the diagnostic weight of chest pain and heart rate.

---

8. Conclusion

This project successfully developed a multi-hospital diagnostic tool for heart disease prediction. By integrating international data and utilizing **5-Fold Cross-Validation**, we have created a model that is robust against the variation in global clinical practices. While the **Random Forest** is the most precise auditor, the **SVM** offers the highest safety margin for preliminary screening. These results demonstrate that machine learning is a powerful decision-support tool in the field of cardiovascular health.

Appendix: Optimal Clinical Configurations

- **Random Forest:** `{criterion: 'gini', max_depth: None, n_estimators: 100}`.
- **SVM:** `{C: 0.1, gamma: 'scale', kernel: 'rbf'}`.
- **KNN:** `{metric: 'manhattan', n_neighbors: 19, weights: 'uniform'}`.

Mathematical Audit Formulas

To rigorously evaluate the clinical performance of the diagnostic models, the following mathematical metrics were utilized:

- **Accuracy:** The ratio of correctly predicted observations to the total observations.
- **Recall (Sensitivity):** The ability of the model to identify all actual diseased cases, which is the primary metric for patient safety in cardiac screening.
- **F_1-Score:** The harmonic mean of Precision and Recall, providing a balanced measure of the model's performance on the diseased class.
- **Matthews Correlation Coefficient (MCC):** A comprehensive metric that accounts for all four quadrants of the confusion matrix, widely considered the most reliable "single number" metric for clinical binary classification.

Clinical Metric Key:

- **TP (True Positive):** Sick patients correctly identified as having heart disease.
- **TN (True Negative):** Healthy patients correctly identified as healthy.
- **FP (False Positive):** Healthy patients incorrectly identified as sick.
- **FN (False Negative):** Sick patients incorrectly identified as healthy; this is the most critical metric to minimize in cardiac diagnostics.